MIL-S-45894(PA)
25 November 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

SWITCH, SENSITIVE: M22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of multipole, double-throw, plunger-actuated switch, which is designated Switch, Sensitive: M22. This switch is mounted on the aft bulkhead of the Pershing warhead section and is used to sense physical separation of the warhead section from the remainder of the missile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC NUOR

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MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

PICATINNY ARSENAL

8861979 - Carton, Packing, Ammunition, for Switch, Sensitive.
10560445 - Switch, Sensitive: M22.

Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence. See 6.2) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly.- The switch shall comply with all requirements specified on drawing (dwg.) 10560445, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article inspection.- This specification contains technical provisions for first article inspection (see 4.3). Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Continuity.- The switch contacts shall exhibit continuity as measured between terminals of the P1 connector which are listed in Table I, and the resistance shall be a maximum of 0.5 ohm. Testing shall be as specified in 4.5.1.

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TABLE I

Plunger Position

Normal		Depressed
A-C	D-F	A-B
G-J	K-L	D-E
K-M		G-H

3.5 Insulation resistance.- The insulation resistance of the switch shall be not less than 50 megohms between the terminals of the P1 connector which are listed in Table II, and the polarity of the test voltage shall be as indicated in Table II. After humidity testing, the insulation resistance of the switch shall be not less than 20 megohms. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.5.2.

TABLE II

Normal Plunger Position

From (+)	To (-)
K	Connector shell and all other pins except M and L
G	Connector shell and all other pins except J
D	Connector shell and all other pins except F
A	Connector shell and all other pins except C

Depressed Plunger Position

From (+)	To (-)
K	Connector shell and all other pins
G	Connector shell and all other pins except H
D	Connector shell and all other pins except E
A	Connector shell and all other pins except B

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3.6 Low temperature.- The switch shall withstand a low temperature of -33° ± 3°F., for a minimum period of 4 hours, without evidence of cracking, distortion, or other damage that may impair its intended operation. While at the low temperature (at the end of the low-temperature exposure period), the switch shall comply with the requirements of 3.4 and 3.5. After this test (at room temperature), the switch shall comply with the requirements of 3.4 and 3.5. Testing shall be as specified in 4.5.3.

3.7 High temperature.- The switch shall withstand a high temperature of 160° ± 5°F., for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. While at the high temperature (at the end of the high-temperature exposure period), the switch shall comply with the requirements of 3.4 and 3.5 After this test (at room temperature), the switch shall comply with the requirements of 3.4 and 3.5. Testing shall be as specified in 4.5.4.

3.8 Temperature shock.- The switch shall withstand alternate exposures to temperatures of 160° ± 5°F. and -65° ± 5°F., for a minimum period of 4 hours at each exposure, and there shall be no evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. After this test (at room temperature), the switch shall comply with the requirements of 3.4 and 3.5. Testing shall be as specified in 4.5.5.

3.9 Humidity.- The switch shall withstand a relative humidity of 95 ± 5 percent and a temperature of 105° ± 5°F. without evidence of deterioration or damage that may impair its intended operation. After this test (at room temperature), the switch shall comply with the requirements of 3.4 and 3.5. Testing shall be as specified in 4.5.6.

3.10 Shock.- The switch shall withstand 18 shocks of 20 ± 2 gravities (g) peak amplitude (sinusoidal) with a shock duration of 11 ± 1 millisecond as measured between points which are 10 percent of the shock amplitude. The switch shall show no evidence of deterioration or damage that may impair its intended operation. During this test, the switch shall comply with the requirements of 3.12. After this test the switch shall comply with the requirements of 3.4 and 3.5. Testing shall be as specified in 4.5.7.

3.11 Vibration.- The switch shall withstand vibrations as specified in Table III without evidence of deterioration or damage that may impair its intended operation. During this test, the switch shall comply with the requirements of 3.12. After this test, the switch shall comply with the requirements of 3.4 and 3.5. Testing shall be as specified in 4.5.8.

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TABLE III

Frequency Range (hertz)	Displacement (D.A.) or Peak g Level (± 10% Tol.)	Minimum Time Period to Cover Freq. Range (minutes)
10-30	0.20 inch	2.0
30-1000	2 g	2.0
1000-2000	2 g	2.0

3.12 Contact chatter.- The switch, with the plunger in the depressed position, shall be monitored for contact chatter during shock and during vibration testing. The following sets of closed contacts shall be connected in series and shall be monitored for contact chatter: A-B, D-E, and G-H. There shall be no discontinuity or chatter in excess of 10 micro-seconds. Testing shall be as specified in 4.5.9.

3.13 Serialization.- A serial numben shall be assigned by the contractor to each sensitive switch produced. In no event, shall serial numbers be duplicated. Six-letter serial codes shall not be used on these switches.

3.14 Soldering.- All soldering shall be in accordance with MIL-STD-1460.

3.15 Workmanship.- All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission.- When the contract requires first article inspection, the contractor shall submit 3 consecutively produced sensitive switches to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection and testing to be performed.- The first article inspection samples shall be inspected for the defects specified in 4.4.2.1, using inspection methods contained therein, and for all requirements of Section 3 (also see Table IV) using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The switches shall be tested in the order specified in Table IV. If any switch fails to comply with any of the applicable requirements, the first article samples shall be rejected. This series of tests is considered destructive.

TABLE IV

	Test	Requirement Paragraph
(a)	Continuity	3.4
(b)	Insulation resistance	3.5
(c)	Low temperature	3.6
(d)	High temperature	3.7
(e)	Temperature shock	3.8
(f)	Humidity	3.9
(g)	Shock	3.10
(h)	Vibration	3.11

4.4 Quality conformance inspection.

4.4.1 Inspection lot.- A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

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4.4.2 Product inspection examination.- A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table V). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

TABLE V

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	29	31-45	27
46-60	35	46-60	30
61-75	38	61-75	33
76-95	42	76-95	35
96-130	46	96-130	37
131-180	50	131-180	39
181-250	52	181-250	41
251-400	54	251-400	42
401 and over	56	401 and over	44

4.4.2.1 Classification of defects.- Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Switch, Sensitive: M22 (see dwg. 10560445).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cable length, min	Gage
	102. Ability of connector to engage and disengage with mating connector	Manual/Gage
	103. Weight, max	Approved Weighing Device
	104. Threads damaged, preventing assembly or disassembly of mounting nuts	Visual/Manual
	105. Connector pin missing or damaged (12 pins)	Visual

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106. Height of pins in connector (see 6.3).....Visual
107. Hardware, on threaded portion of switch, missing or improper.....Visual
108. Keyway, on threaded portion of switch, missing or improper.....Visual
109. Soldering improper (inspect during assembly of connector) (see 3.14).....Visual
110. Switch damaged (including gouges and punctures).....Visual

Minor:
201. Connector potting missing or improper.....Visual
202. Cable sleeving cut or damaged exposing inner conductors.....Visual
203. Connector insert material cut or damaged.....Visual
204. Protective cap on connector missing or improper.....Visual
205. Marking missing, illegible, or improper.....Visual
206. Evidence of poor workmanship (see 3.15).....Visual

4.4.3 Inspection testing.- The contractor shall perform all of the tests specified in this paragraph. The tests shall be performed in the following sequence:

	Test	Test Classification
(a)	Continuity	Major
(b)	Insulation resistance	Major

None of these tests is considered destructive, and samples so tested shall be returned to the lot.

4.4.3.1 Continuity (see 3.4) and insulation resistance (see 3.5).- Each sensitive switch shall be subjected to these tests (see applicable paragraph in 4.5). If any switch fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

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4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests.- Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions designed or specified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration instructions prior to use on the contract (see 6.4).

4.5 Test methods.- The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F.). NOTE: (1) During environmental tests, the switch's connector shall be protected by a mating connector or by a suitable protective cap. (2) The hardware, on the threaded portion of the switch, may be removed for any test. (3) During the low- and high-temperature tests (see 4.5.3 and 4.5.4), the switch may be removed from the temperature chamber for testing at the end of the low- or high-temperature exposure period; however, the exposure to room temperature shall not exceed 5 minutes. The switch shall be replaced in the temperature chamber for a minimum period of 20 minutes before each subsequent 5-minute measuring period.

4.5.1 Continuity.- The continuity of the switch contacts shall be measured with a Government-approved continuity tester, and the test current shall be limited to 100 milliamperes (ma). The switch shall comply with the requirements of 3.4 during this test.

4.5.2 Insulation resistance.- The insulation resistance of the switch shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 500 ± 50 volts direct current. The test voltage shall be applied instantaneously to the switch, and the insulation resistance shall be measured after a 10-second application of this test voltage. The switch shall comply with the requirements of 3.5 during this test.

4.5.3 Low temperature.- The switch shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The switch shall remain at this temperature for a minimum period of 4 hours and then tested as specified in 3.4 and 3.5 at the end of the low-temperature exposure period. The switch shall then be removed from the test chamber, allowed to stabilize at room temperature, and then tested for the requirements of 3.4 and 3.5. The switch shall comply with the requirements of 3.6 during this test.

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4.5.4 High temperature.- The switch shall be placed into a test chamber, and the temperature of the chamber shall be raised to 160°F. The switch shall remain at this temperature for a minimum period of 4 hours and then tested as specified in 3.4 and 3.5 at the end of the high-temperature exposure period. The switch shall then be removed from the test chamber, allowed to stabilize at room temperature, and then tested for the requirements of 3.4 and 3.5. The switch shall comply with the requirements of 3.7 during this test.

4.5.5 Temperature shock.- The switch shall be placed into a test chamber having an internal temperature of 160°F., and it shall remain at this temperature for a minimum period of 4 hours. At the end of this test exposure, and within a 5-minute period, the switch shall be transferred to a test chamber having an internal temperature of -65°F., and it shall remain at this temperature for a minimum period of 4 hours. This high- and low-temperature exposure constitutes one cycle, and the switch shall be subjected to three such cycles. At the completion of the third cycle (at the end of the low-temperature exposure period) the switch shall be removed from the test chamber, allowed to stabilize at room temperature, and then tested for the requirements of 3.4 and 3.5. The switch shall comply with the requirements of 3.8 during this test.

NOTE: The test may, for convenience purposes, be interrupted at the end of a low-temperature exposure period. This is accomplished by returning the switch to room temperature after an exposure at high temperature of 1 hour minimum. To continue the test, start with the high-temperature exposure of the test cycle.

4.5.6 Humidity.- The switch shall be placed into a dry oven and conditioned at a temperature of 105°F. for a minimum period of 24 hours. At the end of this conditioning period, the switch shall be transferred to a humidity chamber having a temperature of 105°F. and a relative humidity of 95 percent. The switch shall remain in this chamber for a minimum period of 96 hours. Following this exposure period, the switch shall be removed from the humidity chamber, allowed to stabilize at room temperature for a minimum period of 4 hours, and then tested for the requirements of 3.4 and 3.5. The switch shall comply with the requirements of 3.9 during this test.

4.5.7 Shock.- The switch shall be securely mounted to a shock testing machine (the cable and connector shall be attached to the shock testing machine by any convenient method and may be cushioned to prevent damage). Applicable

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switch contacts shall be monitored for contact chatter throughout this test (see 3.12). The switch shall be shocked three times, in both directions along each of three mutually perpendicular axes, for a total of 18 shocks. One of the three mutually perpendicular axes shall correspond to an axis parallel to the plunger of the switch. The switch shall comply with the requirements of 3.10 during this test.

4.5.8 Vibration.- The switch shall be securely mounted to a vibration machine (the cable and connector shall be attached to the vibration machine by any convenient method and may be cushioned to prevent damage). Applicable switch contacts shall be monitored for contact chatter throughout this test (see 3.12). The switch shall be subjected to the vibrations specified in Table III along each of three mutually perpendicular axes. One of the three mutually perpendicular axes shall correspond to an axis parallel to the plunger of the switch. The switch shall comply with the requirements of 3.11 during this test.

4.5.9 Contact chatter.- A suitable end cap shall be secured to the switch in order to keep the plunger depressed during this test (any other means of keeping the plunger depressed may be used). A suitable low-voltage, direct-current power supply shall be used to supply a low current (maximum of 100 ma) through the sets of contacts, which are connected in series (see 3.12), and through a Government-approved contact chatter detector. The switch shall comply with the requirements of 3.12 during this test.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking shall be in accordance with FED-STD-356 or as specified in the contract or purchase order.

5.2 Spare parts.

5.2.1 Preservation and packaging.- Preservation and packaging shall be level A or commercial packaging as specified (see 6.1).

5.2.1.1 Level A.- The switch shall be preserved, packaged, and marked in accordance with dwg. 8861979.

5.2.1.2 Commercial packaging.- The switch shall be preserved, packaged, and marked in accordance with FED-STD-356.

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5.2.2 Packing and marking.- The switch shall be packed and marked in accordance with FED-STD-356 or as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

6.1.1 Procurement requirements:
(a) Title, number, and date of this specification.
(b) Production item or spare part procurement.
(c) Selection of method of preservation and packaging for spare parts procurement (see 5.2.1).

6.1.2 Contract data requirements.- Contractor engineering documentation, specified in 4.4.4.1, will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

6.3 Inspection of connectors.

6.3.1 Height of pins.- In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.040 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

6.4 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SARPA-QA-T.

6.5 Supersession data.- This specification supersedes the requirements of Purchase Description PA-PD-2453, dated 4 May 1964, with Amendment 1 dated 20 July 1973.

Custodian:
Army-MU

Project No.: NUOR-A062

Preparing Activity:
Army-MU

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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